

Chapter 66: Neoplastic Disorders — Revision Notes

Williams Obstetrics 26e, pp. 3025–3076 of the PDF. High-yield numbers are in **bold**. The three tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 6 figures are included from the PDF.

Big picture: Most neoplasms in pregnancy are **benign** (leiomyomas and ovarian cysts are the most frequent). Malignancy complicates **~1 in 1000–2000 pregnancies**; **one third** are diagnosed prenatally and the rest within 12 months of delivery. Treatment is individualized: cancer type and stage, desire to continue the pregnancy, and the risk of delaying or modifying treatment.

- Cancer Research Network: **breast 25%, thyroid 20%**; melanoma, hematological and cervical ~10% each.
- Cancer in pregnancy is associated with adverse pregnancy outcomes in general.

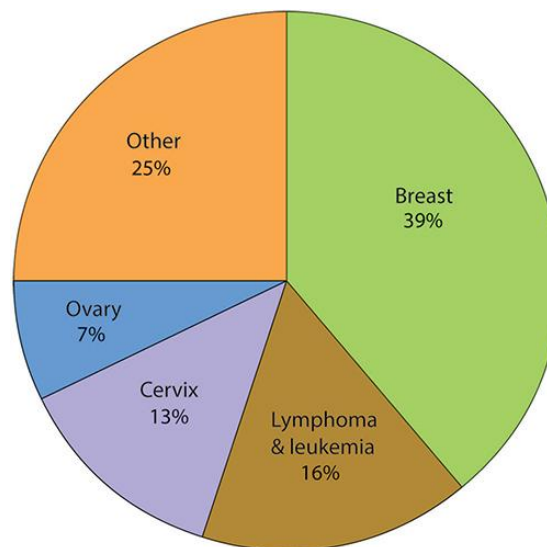


Figure 66-1. Cancers in 1170 pregnant women (INCIP): breast 39%, lymphoma/leukemia 16%, cervix 13%, ovary 7%, other 25%.

1. Cancer Therapy in Pregnancy

Surgery

- Nonreproductive-tract surgery is **well tolerated** by mother and fetus.
- Deferring surgery until **12–14 wk** (miscarriage fear) is probably unnecessary. **Operate at any GA if maternal well-being is imperiled.**
- **Pregnancy + cancer = ↑ VTE** (antepartum and postpartum). No specific guidelines → consider **prophylactic LMWH + stockings and/or pneumatic compression**, depending on the procedure.

Diagnostic imaging

- **Sonography preferred.** Most diagnostic x-rays give very low doses and **should not be delayed** if they would change therapy (ACOG).
- **MR imaging: safe in any trimester** (delaying past the 1st trimester may ↓ risk).
- ⚠ **Gadolinium: not in the 1st trimester**; later only if benefit overwhelmingly outweighs risk.
- **CT (ionizing):** mainly for acute problems (PE, bowel/renal obstruction, acute neurological events). Oral and IV contrast have no known fetal harm; **no need to interrupt breastfeeding.**

Radiation therapy

- Fetal risk depends on **dose, tumor location, field size and GA**. Effects: malformation, **intellectual disability**, FGR, sterility, carcinogenesis.

Gestational period	Main effect	Threshold
First 2 wk after fertilization	Chromosomal damage, embryonic death ("all or none")	—
Wk 2–8 (organogenesis)	Malformations	0.1–0.2 Gy
Wk 8–15	CNS: intellectual disability	~0.06 Gy
Wk 16–25	CNS: intellectual disability	~0.25 Gy
>25 wk	Less susceptible, but no GA is safe for therapeutic doses	—

- ⚠ **Radiotherapy to the maternal abdomen is contraindicated.** Supradiaphragmatic fields (some head and neck cancers) are relatively safe **with abdominal shielding**.

Chemotherapy

- Fetal concerns: malformations, FGR, intellectual disability, future childhood cancer. Risk depends mostly on **GA at exposure**.
- 1st trimester (organogenesis) = most harmful:** 14% of major malformations were attributable to 1st-trimester cytotoxic exposure.
- After the 1st trimester, most agents have no immediate obvious fetal harm;** late mutagenic effects appear limited.
- Withhold chemotherapy in the 3 weeks before expected delivery** (maternal neutropenia/pancytopenia → infection, hemorrhage; limited neonatal drug clearance).
- Most cytotoxic drugs are contraindicated with breastfeeding.**

Molecular and targeted therapy

Agent	Pregnancy data
G-CSF (filgrastim, pegfilgrastim)	Limited data support safety
Erythropoietin alfa	Appears safe; risk of maternal hypertension
Romiplostim, eltrombopag (TPO-receptor agonists)	Appear safe
Tyrosine kinase inhibitors (most are FDA class D)	Embryotoxic/teratogenic in the 1st trimester → balance benefit and risk
Trastuzumab (anti-HER2)	Not teratogenic, but 2nd–3rd trimester use → oligohydramnios (reversible on stopping)
Other HER2 inhibitors	Sparse data → avoid

- Targeted therapy = **monoclonal antibodies** and **small-molecule inhibitors**; also **nanotherapy** (drugs in encapsulated nanoparticles).

Fertility and pregnancy after cancer therapy

- Oncofertility:** counsel **before** treatment (ASRM guidelines).
 - Embryo or oocyte cryopreservation** = recognized option. **Ovarian tissue cryopreservation** in research settings.
 - Many conceive with **ART** (which has its own obstetrical risks).
- Ovarian transposition** before pelvic radiation: ovary moved on its blood supply and fixed to the lateral abdominal wall **3–4 cm above the umbilicus**, marked with clips → function preserved in **65–94%**.
- Offspring of survivors: most chemo/radiation does **not** ↑ congenital anomalies or genetic disease.

- **Childhood chemotherapy:** no consistent link with adverse obstetrical outcome. Adult cancer: slightly ↑ preterm birth and cesarean.
- **Prior abdominopelvic radiation** → ↑ **abortion, LBW, stillbirth, preterm birth** (↓ uterine volume, thin endometrium, ↓ uterine blood flow). Worse if the uterus was irradiated at a **younger age**.

Placental metastases

- Rare. Commonest: **melanoma**, leukemias, lymphomas, **breast cancer**.
- **Send every placenta from a woman with cancer for histology.** Tumor cells usually stay in the **intervillous space**; **fetal metastases are rare**.

Memory hook: Placental metastasis = "My Little Lovely Baby" (Melanoma, Leukemia, Lymphoma, Breast).

2. Reproductive Tract Neoplasms

- Benign: leiomyomas, ovarian tumors, endocervical polyps. Of reproductive-tract cancers in pregnancy, **cervical neoplasia is the most common**.

Cervix

Endocervical polyp

- Overgrowth of endocervical stroma; single red, elongated, fleshy mass from the canal. Can bleed; can cause **AGUS** on Pap.
- On histology: **dysplasia up to 0.5%, cancer ~0.1%**.
- Small asymptomatic polyps can be left to slough at delivery. Remove if **malignancy suspected or bleeding troublesome**: grasp with ring forceps and **twist repeatedly** until the base avulses; **Monsel paste** (ferric subsulfate) for hemostasis. Thick pedicle → ligation and excision.

Cervical neoplasia screening

- Pregnancy = good time to screen. **Note pregnancy on the requisition** (decidual cells, **Arias-Stella reaction** can mimic atypical glandular cells).
- **Same screening guidelines as nonpregnant (average risk):**

Age	Screening
<21	None
21–29	Cytology alone every 3 yr
≥30	Co-testing (HPV + cytology) every 5 yr, primary HPV every 5 yr, or cytology every 3 yr

- **Higher risk** (screen earlier and more often, per HIV guidelines): **HIV**, SLE, transplant, immunosuppressants, prior CIN or early cervical cancer, **in utero DES**.
 - HIV, **≤30 yr**: Pap at diagnosis, then **yearly**; after **3 consecutive normal** → every 3 yr. **No co-testing**.
 - HIV, **>30 yr**: same Pap schedule; co-testing allowed; if negative, repeat in **3 yr**.

Human papillomavirus

- Prevalence in pregnancy **~15%**. Usually clears.
- **HPV 16 and 18** → HSIL and cancer; detection prompts **colposcopy**. HPV testing targets **high-risk types only**; **no role for low-risk testing**.
- **HPV 6 and 11** → **genital warts**. Neonatal conjunctival, **laryngeal**, vulvar or perianal warts at birth or within **1–3 yr** are probably from perinatal exposure.
- ⚠️ **Cesarean does not ↓ neonatal laryngeal papillomatosis**.
- **Gardasil 9 (9vHPV)**: types 6, 11, 16, 18, 31, 33, 45, 52, 58. Not given in pregnancy; compatible with breastfeeding.

The extracted chapter text lists HPV type "35" in Gardasil 9; the vaccine covers type 33.

Abnormal cytology and histology

- Incidence and management are the same as in nonpregnant women. **ASCCP 2019** guidelines are **risk-based** (risk of CIN 3+), using online algorithms.
- **Goal in pregnancy = exclude invasive cancer.** Biopsy lesions suspicious for high-grade disease or cancer.
- **⚠ Unacceptable in pregnancy: endocervical sampling (ECC), endometrial biopsy, treatment of preinvasive lesions.**
- **CIN on biopsy → vaginal delivery allowed;** re-evaluate **≥4 weeks postpartum**. Higher-risk findings → repeat colposcopy + cytology/HPV **no more often than every 12 weeks**; rebiopsy only if worse or invasion suspected.
- **Regression is common:**

Study	Finding
1079 dysplasias	61% normal postpartum
CIN 2–3	70% regressed; 7% of CIN 2 → CIN 3; none → invasive
77 CIS	$\frac{1}{3}$ regressed, $\frac{2}{3}$ persisted; only 2 microinvasive on postpartum cone

- **AIS** is managed like CIN 3: **no treatment until 6 weeks postpartum** unless invasive; refer to an expert colposcopist.

Cervical conization

- **Only if invasion is suspected** (LEEP or cold knife). Can't excise deeply into the canal (membrane rupture) → **residual disease in 43%**; **~10%** needed transfusion.
- **⚠ Avoid in pregnancy** if possible: abortion, membrane rupture, **hemorrhage**, preterm delivery.
- **Prior treatment complications:**
 - **Cervical stenosis** (after cone, LEEP, laser): almost always yields in labor. **"Conglutinated cervix"** effaces without dilating → **fingertip pressure**; rarely instrument dilation or cruciate incisions.
 - **Preconceptional cold-knife cone → cervical insufficiency and preterm birth.** LEEP–preterm link debated; risk $\propto$ **size of tissue excised**.

Invasive cervical cancer

- **1 in 1200–10,000 pregnancies.** **Squamous 75%**, adenocarcinoma the rest.
- Exophytic/endophytic, polyp, papillary, **barrel-shaped cervix**, ulceration/necrosis; watery, purulent, foul or bloody discharge.
- **Single Pap has poor sensitivity → biopsy suspicious lesions** (Tischler forceps). Bleeding → Monsel paste + pressure.
- **Staging** now includes **imaging, surgical findings and lymph nodes**. Clinical components: cold-knife cone, EUA, cystoscopy, IVP (or MR), proctoscopy, CXR.
- **⚠ Pregnancy softening of the parametria/broad ligament base → extent is underestimated.** Use **MR without gadolinium** (urinary tract, parametrium, nodes).

Management and prognosis

Situation	Management
Stage IA1 (microinvasive, <3 mm) on cone	Like intraepithelial disease; continue pregnancy, vaginal delivery , definitive therapy 6 wk postpartum
Deeper invasion, first half of pregnancy	Immediate treatment (with the decision about continuing pregnancy)
After 20 wk, stage I–IIA, non-bulky	Delayed treatment reasonable
Option	Laparoscopic lymphadenectomy; delay if nodes negative. Neoadjuvant platinum chemotherapy promising
Stage I and early IIA (young women)	Radical hysterectomy + pelvic lymphadenectomy preferred (surgery = radiation in efficacy)
— before 20 wk	Radical hysterectomy with fetus in situ
— later pregnancy	Hysterotomy/delivery first , then radical hysterectomy
Advanced stage	Radiotherapy : early → spontaneous abortion (curettage if not); 2nd trimester → hysterotomy in up to ¼ (induction/D&E risk hemorrhage)

- **Radiation vs surgery (stage IB): severe complications 30% vs 7%.** Radiation destroys ovarian and possibly sexual function; bowel/urinary injury.
- Rare options: **abdominal radical trachelectomy <20 wk (IB1)**; **laser conization** for IA1 adenocarcinoma.
- **Pregnancy does not worsen prognosis; 5-yr survival ~80%** (pregnant = nonpregnant).
- **Delivery**: effect of vaginal birth through a cancerous cervix unknown; hemorrhage from bulky/friable tumors and **recurrence in the episiotomy scar** → **most favor cesarean**.

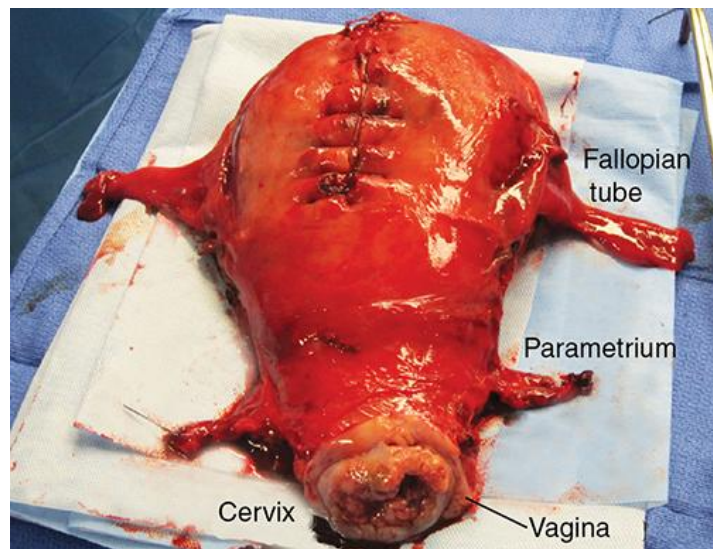


Figure 66-2. Radical hysterectomy specimen: parametrium and 2–3 cm of vagina are removed for a tumor-free margin.

Pregnancy after radical trachelectomy

- Fertility-sparing for **stage IB1–IB2**: cervix amputated at the internal os, **permanent isthmic cerclage**, isthmus joined to the vagina.
- ⚠️ **Classical cesarean required** (permanent cerclage).
- **Cervical length <13 mm at 20 wk** → **preterm birth <34 wk**.
- 2566 women: **5-yr survival 97%**; pregnancy rate **24%**; live-birth rate **75%**.

Uterus

Leiomyomas

- Benign smooth-muscle tumors; **~2% of pregnancies**. 1st-trimester prevalence **18% in Black vs 8% in White** women.

- Submucosal, subserosal, intramural; less often cervical or broad ligament. Rare: **parasitic** (omental blood supply), **leiomyomatosis peritonealis disseminata** (mimics carcinomatosis; often regresses after pregnancy).
- **Progesterone is the critical mitogen**; estrogen maintains progesterone receptors. In pregnancy, growth is **unpredictable** (grow, shrink or stay the same).
- **Sonography is indispensable**; no routine surveillance unless complications are expected. MR (after the 1st trimester) if unclear.

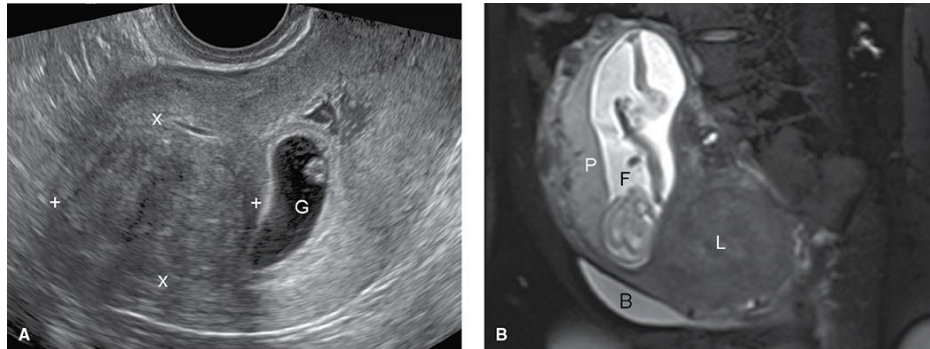


Figure 66-3. Leiomyoma imaging: (A) TVS fundal myoma beside an early sac; (B) MR at 16 wk, large posterior lower-segment/cervical myoma filling the pelvis (hydronephrosis; planned cesarean).

Symptoms

- Mostly asymptomatic; pain/pressure with large masses. Chronic pain → non-narcotic analgesics.
- **Red (carneous) degeneration**: myoma outgrows its blood supply → hemorrhagic infarction → **acute focal pain and tenderness**, ± leukocytosis, low-grade fever. **Diagnosis of exclusion**; observe closely. Can trigger preterm labor. **Settles within a few days**.
- **Myomectomy in pregnancy** only in highly selected cases (23 cases, 14–20 wk, half for pain; mostly term cesarean). **Torsed pedunculated subserosal myoma** → ligate the stalk (limit surgery to a **discrete pedicle**).

Pregnancy complications (2065 women)

Complication	Risk vs no myoma
Placental abruption	4×
Breech presentation	4×
1st-trimester bleeding	2×
Dysfunctional labor	2×
Cesarean delivery	6×
Preterm birth (301 women)	2.5×
2nd-trimester abortion	8×

- Key determinants: **number, size and location**. **Placenta over/adjacent to a myoma** → ↑ abortion, preterm labor, abruption, PPH; **retroplacental** → **FGR**.
- **Prolapsed submucous myoma** → ligate the stalk vaginally near term.
- Cervical/lower-segment myomas can **obstruct labor**. Yet **70% delivered vaginally** even with myomas **≥10 cm** → **trial of labor unless the birth canal is clearly obstructed**.
- **At cesarean**: palpate the fundus to exclude **uterine malrotation**. ⚠ **Avoid myomectomy at cesarean** (hemorrhage) unless there is intractable bleeding or the myoma prevents closure. Cesarean hysterectomy is difficult (**lateral ureteral displacement**).
- **Infection** is rare: usually postpartum (near the implantation site), septic abortion, or perforation by an instrument.

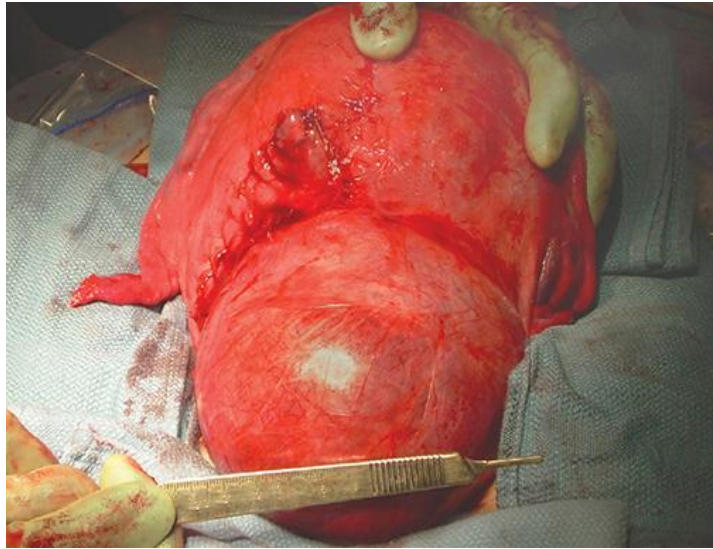


Figure 66-4. Large lower-segment leiomyoma at cesarean: a classical vertical incision (left of the myoma) was needed.

Fertility considerations

- Effect on fertility unclear. Myomectomy for intramural/subserosal myomas: evidence inconclusive. **Hysteroscopic resection of submucous myomas** improves pregnancy rates (ASRM, fair evidence).
- Myomas **not linked to** ↑ miscarriage (≈1400 pregnancies).
- **After myomectomy: uterine rupture before or during labor** → read the operative note. **Cavity entered or defect adjacent to it** → **cesarean before labor**, at **37 0/7–38 6/7 wk** (ACOG).
- **Uterine artery embolization**: ↑ miscarriage, cesarean, PPH; **relatively contraindicated** if future pregnancy is planned (SIR). Does not harm ovarian reserve.

Endometrial lesions

- **Endometriosis**: most consistent association is **placenta previa**.
- **Scar endometrioma** after laparotomy or episiotomy (more with **Pfannenstiel** than vertical): palpable mass with **cyclical pain**.
- **Adenomyosis**: partly linked to curettage disrupting the endometrial–myometrial border. Associated with miscarriage, FGR, **preeclampsia**, previa, preterm birth.
- **Endometrial carcinoma**: estrogen-dependent, usually >40 yr → rare. Of 27 cases, most found in **1st-trimester curettage**. Usually **early, well-differentiated adenocarcinoma** → **TAH-BSO**.
 - Fertility-sparing (well-counseled, early, well-differentiated endometrioid): **oral or intrauterine progestin**. Live-birth rates low, often need ART; recurrences and deaths reported.

Ovary

- Adnexal masses: **1 in 100 to 1 in 2000 pregnancies**. Ovarian malignancy **1 in 19,000** (California).
- **Commonest: corpus luteum cysts, endometriomas, benign cystadenomas, mature cystic teratomas (dermoids)**.
 - Endometriomas in the 1st trimester: ¼ stable, ¼ grow, ½ shrink. Dermoids rarely enlarge.
 - Rare: teratoma neural tissue with **NMDA receptors** → **anti-NMDA-receptor encephalitis**.
- Malignancy uncommon: **1% malignant + 1% low malignant potential** (9375 masses). **4–13%** of surgically excised masses.

Diagnosis

- Mostly asymptomatic; pressure/chronic pain; acute pain from **torsion, rupture, hemorrhage**.
- Found on routine sonography. **MR without contrast** for complex anatomy.
- **CA125**: normally ↑ in **early pregnancy and the early puerperium** (decidua); **2nd trimester to term not higher than nonpregnant**. Abnormally ↑ in **severe preeclampsia**.
- Other markers: hCG, AFP, inhibin A/B, CEA, OVA1.

Torsion

- Acute constant or episodic lower pain, tenderness, **nausea and vomiting**; usually with an adnexal mass.
- **TVS + color Doppler: absent flow strongly correlates** with torsion, but early twisting may block **only venous flow** (arterial flow present).
- Other signs: **stromal edema, peripheral follicles, free fluid, whirlpool sign** (twisted pedicle vessels), **follicular ring sign** (1–2 mm echogenic rim around antral follicles).
- **Suspected torsion → laparoscopy or laparotomy. Untwist to save the ovary** (PE fear unfounded). Remove a **grossly necrotic** adnexum. A salvageable ovary is decongested within minutes but often **stays blue-black**.
- Viable adnexum: resect any neoplasm (cystectomy may be hard in an edematous ovary → adnexectomy; or delay cystectomy for low-risk lesions). **Oophoropexy** (especially for a second torsion): shorten or fix the utero-ovarian ligament to the posterior uterus, pelvic sidewall or round ligament.

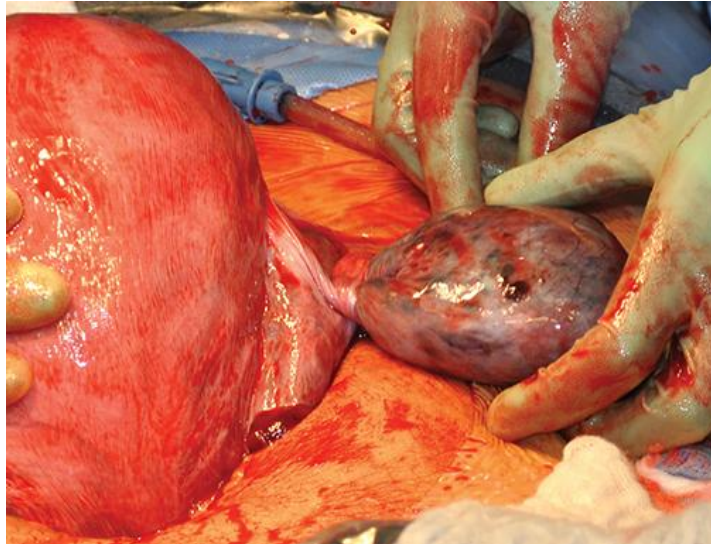


Figure 66-5. Ovarian torsion at term, found at cesarean; the utero-ovarian pedicle was untwisted.

Hemorrhage: ruptured corpus luteum cyst

- Most common cause of ovarian hemorrhage → hemoperitoneum, diffuse lower pain. TVS: blood in the cul-de-sac, collapsed cyst.
- Certain diagnosis + settling symptoms → observe. Ongoing bleeding → surgery.
- **⚠ Corpus luteum removed before 10 wk → progesterone support:**

Regimen	Dose	Duration
Micronized progesterone (Prometrium)	200 or 300 mg orally daily	Until 10 completed wk
8% vaginal progesterone gel (Crinone)	1 applicator daily + oral micronized progesterone 100–200 mg daily	Until 10 completed wk
17-OHP caproate IM	150 mg	8–10 wk: one dose right after surgery. 6–8 wk: two more doses at 1 and 2 wk later

Asymptomatic adnexal mass

Size / features	Management
Cystic, benign-looking, <5 cm	Usually no further surveillance (likely corpus luteum; resolves by early 2nd trimester)
5–10 cm	TVS + color Doppler ± MR. Simple → expectant with sonographic surveillance; resect if it grows, looks malignant or causes symptoms
≥10 cm	Substantial risk of malignancy, torsion, labor obstruction → surgical removal reasonable
Classic endometrioma or dermoid	Resect postpartum or at cesarean
Suspicious for cancer	Prompt resection

- **Malignant features: nodules, papillary excrescences, solid components, thick vascular septa, ascites.**
- 563 masses: half simple (**1% malignant**), half complex (**9% malignant**).
- **~1 in 1000** pregnant women have adnexal surgery. Plan resection at **14–20 wk** (most that will regress have done so). **Laparoscopy is ideal.**
- **Refer to a gynecologic oncologist (ACOG):** very high CA125, **nodular or fixed mass, ascites, metastases**, or malignant sonographic findings.

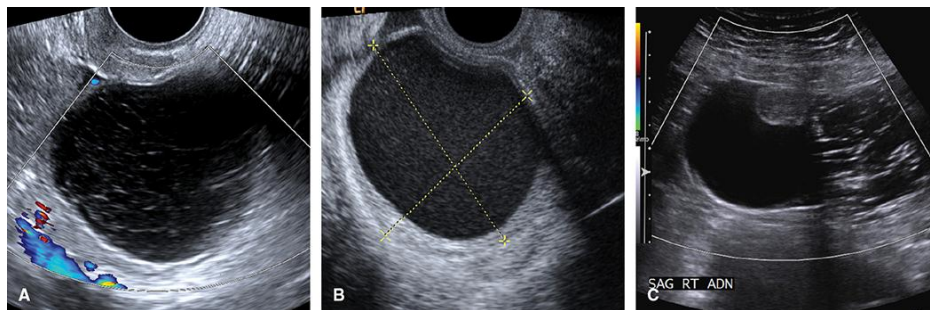


Figure 66-6. Adnexal masses: (A) corpus luteum with reticular pattern; (B) homogeneous low-level echoes = endometrioma or hemorrhagic corpus luteum; (C) dermoid with hair lines/dots, Rokitansky nodule and fat–fluid level.

Pregnancy-related ovarian tumors

Feature	Pregnancy luteoma	Hyperreactio luteinalis	Ovarian hyperstimulation syndrome
Nature	Rare benign tumor of luteinized stromal cells	Multiple large theca-lutein cysts , uni- or bilateral, usually after the 1st trimester	Multiple follicular cysts + ↑ capillary permeability
Cause	Classically ↑ testosterone	Very high hCG: GTD, twins, hydrops, ↑ placental mass	Ovulation induction (rarely spontaneous); hCG → VEGF
Imaging	Solid , may be multiple/bilateral, complex if hemorrhagic	"Spoke-wheel" pattern	—
Maternal virilization	Up to 25%	Possible	Absent
Fetal virilization	~Half of female fetuses of virilized mothers (placental aromatization protects most)	No	No
Management	No surgery unless torsion, rupture, hemorrhage; regresses postpartum ; lactation delayed ~1 wk; rarely recurs	No surgery if confident and uncomplicated; resolves after delivery	Supportive: maintain volume, prevent VTE ; paracentesis if severe

- **Luteoma differential:** granulosa cell tumor, thecoma, Sertoli-Leydig, Leydig cell tumor, stromal hyperthecosis, hyperreactio luteinalis. Normal pregnancy testosterone can be quite high.
- Hyperreactio: **testosterone does not predict** maternal virilization.
- **OHSS complications:** ascites, pleural/pericardial effusion, **hypovolemia with AKI, hypercoagulability**, ARDS, ovarian rupture, VTE.

Ovarian cancer

- **Leading cause of death from genital-tract cancer** in all women, but in pregnancy only **1 in 20,000–50,000** births.
- **75% early stage** → **5-yr survival 70–90%**.
- **Types in pregnancy (decreasing order): germ cell and sex cord–stromal > low malignant potential > epithelial.**
- Pregnancy does **not** alter prognosis. Manage as in nonpregnant women, modified for GA.
- **Frozen section confirms malignancy** → **surgical staging:** washings, diaphragm and peritoneal biopsies, **omentectomy**, pelvic and infrarenal para-aortic nodes if accessible.
- Advanced disease: bilateral adnexectomy + omentectomy (debulking). Early pregnancy: hysterectomy and aggressive debulking may be chosen. **Chemotherapy in pregnancy** while awaiting lung maturity for aggressive disease.
- **⚠ CA125 is not accurate for monitoring chemotherapy in pregnancy.**
- **Protective: parity**, prior miscarriage/abortion, **COCs, breastfeeding**, tubal interruption/salpingectomy.

Adnexal (paratubal/paraovarian) cysts

- Paramesonephric remnants or mesothelial inclusion cysts; **5%** at autopsy.
- **Hydatid of Morgagni** = commonest; pedunculated, hangs from the fimbria. Can twist. Usually found at cesarean or sterilization → **excise or window-drain**.
- Neoplastic paraovarian cysts are rare, rarely borderline/malignant.

Vulva and vagina

- **VIN and VAIN** (HPV-related) are more common than invasive disease → **treat after delivery**.
- Vulvar/vaginal cancer is rare in pregnancy; **biopsy any suspicious lesion**. Mostly **squamous**.
- **Radical surgery for stage I is feasible even in the 3rd trimester**. Vaginal delivery OK if incisions are well healed. Late pregnancy: definitive therapy can often wait (slow growth).

3. Breast Carcinoma

- **Most frequent cancer complicating pregnancy; ~1 in 15,000 births**. Rising with **delayed childbearing**.

Diagnosis

- **>90% have a palpable mass; >80% self-detected**. Diagnosis often **delayed** (pregnancy breast growth obscures masses).
- **Evaluation = same as nonpregnant**. Pursue any suspicious mass to diagnosis; biopsy or excise a discrete palpable mass.
- **Sonography:** high sensitivity/specificity for solid vs cystic.
- **Mammography** allowed: fetal dose **0.04 mGy** with shielding, but **false-negative 35–40%** (dense breasts).
- **MR imaging** if the decision to biopsy is uncertain.

Table 66-1. American College of Radiology: Breast Imaging Reporting and Data System — BI-RADS^a

BI-RADS Category	Interpretation	Examples
0	Additional mammography views or sonography required	Focal asymmetry, microcalcifications, or a mass identified on a screening mammogram
1	No abnormalities identified	Normal fat and fibroglandular tissue
2	Not entirely normal, but definitely benign	Fat necrosis from a prior excision, stable biopsy-proven fibroadenoma, stable cyst
3	Probably benign, <2% probability of malignancy	Circumscribed mass that has been followed for <2 years
4A	Low suspicion for malignancy (2–9%), but intervention required	Probable fibroadenoma, complicated cyst
4B	Moderate suspicion for malignancy (10–49%), intervention required	Partially indistinctly margined mass otherwise consistent with a fibroadenoma
4C	High suspicion (50–94%), but not classical for carcinoma	New cluster of fine pleomorphic calcifications, ill-defined irregular solid mass
5	Almost certainly malignant (>95% probability)	Spiculated mass, fine linear and branching calcifications
6	Biopsy-proven carcinoma	Biopsy-proven carcinoma

^aMammography, sonography, or magnetic resonance imaging. The printed third column has no heading; "Examples" is added here. Category 0 is printed as "Addition mammography views", corrected to "Additional".

Cysts and solid masses

Lesion	Features	Management
Simple cyst	—	No special management ; aspirate if symptomatic
Complicated cyst	Internal echoes; may look solid	Aspirate ; incomplete resolution → core-needle biopsy
Complex cyst	Septa or intracystic mass	Excise
Solid mass	Triple test : clinical exam + imaging + core-needle biopsy	Concordant benign (≥99% accurate) → follow clinically. Any one suspicious → excise

Management

- **Metastatic workup (limited): CXR, liver sonography, skeletal MR**; MR can also image abdomen, pelvis, brain.
- Team: obstetrician, breast surgeon, medical oncologist.
- **Terminating the pregnancy does not change prognosis.**
- Treatment mirrors nonpregnant care, **except**:
 - **Chemotherapy and surgery → postponed to the 2nd trimester**
 - **Adjuvant radiotherapy → after delivery**
- **⚠ Avoid delay**: metastasis risk ↑ **5–10% every few months**.
- **Surgery may be definitive**: wide excision or modified/total mastectomy, each **with axillary staging**. **Sentinel node biopsy with Tc-99m is safe**. Reconstruction usually after delivery (immediate reconstruction reported in 10 women).
- **Chemotherapy** for node-positive and node-negative disease (improves premenopausal survival). Multiagent if delivery is not within several weeks: **cyclophosphamide, doxorubicin, cisplatin. Anthracycline → pretreatment maternal echocardiography** (cardiotoxicity).
- **⚠ Trastuzumab is not recommended**: HER2/neu (in ~1/3 of invasive cancers) is strongly expressed in **fetal renal epithelium** → miscarriage, **fetal renal failure/oligohydramnios**, preterm birth.

Prognosis

- More aggressive in young women; whether more aggressive in pregnancy is debatable.
- **Stage for stage, 5-yr survival is comparable**, but pregnant women present **later (up to 60% node-positive at diagnosis)** → overall worse prognosis.

Pregnancy following breast cancer

- Survival **not adversely affected**; pregnancy **≥10 months after diagnosis** of early cancer may even confer a survival benefit. Breastfeeding does not alter the course.
- **Delay conception 2–3 years** (recurrences cluster early).
- **⚠ Hormonal contraception is contraindicated** → **copper IUD** is excellent.
- Breast cancer within 10 years of delivery → ↑ metastasis risk.
- **⚠ Tamoxifen is teratogenic** → **stop 2 months before conception**.

BRCA1/2 mutations — hereditary breast and ovarian cancer (HBOC)

Table 66-2. Some Genes Associated with Hereditary Cancers

Cancer	Genes with established association
Breast	BRCA1, BRCA2, STK11, CDH1, PALB2, CHEK2, ATM, TP53, PTEN, NF1, NBN
Epithelial ovarian^a	BRCA1, BRCA2, BRIP1, MSH2, MSH6, MLH1, RAD51C, RAD51D
Uterine	MSH2, MSH6, MLH1, PMS2

^aSTK11 mutation is associated with ovarian sex-cord stromal tumors but not epithelial types. The printed table reads "BRIP1"; the gene is BRIP1.

- **Autosomal dominant.**
- **Ovarian screening (TVS/CA125): not routinely recommended.**
- **Breast surveillance:**
 - **25–29 yr:** clinical breast exam **1–2×/yr** + **annual breast MR**
 - **Older:** annual MR and annual mammography, **alternating every 6 months**
- **Chemoprevention: COCs ↓ ovarian cancer** (breast effect conflicting: none or modest ↑). **Tamoxifen ↓ breast cancer in BRCA2.**
- **Risk-reducing BSO after childbearing: BRCA1 at 35–40 yr; BRCA2 at 40–45 yr.** BSO or mastectomy also ↓ breast cancer.
- **Preimplantation genetic testing** of blastocysts is an option. **Prenatal testing** for adult-onset conditions is **discouraged by ACOG** (ethical concerns).

Factor	BRCA1	BRCA2
Parity → breast cancer	Conflicting	Conflicting
Parity → ovarian cancer	↓	No effect
Breastfeeding → breast cancer	↓	No effect
Breastfeeding → ovarian cancer	↓	↓
Induced abortion → breast cancer	Not ↑	Not ↑

- Breast cancer diagnosed in pregnancy does not lower survival in BRCA carriers. Pregnancy after treatment → no ↑ complications or worse survival.

Memory hook: "**BRCA1** = **1**st to BSO (35–40); **BRCA2** = **2**nd (40–45)."

4. Thyroid Cancer

- Palpable nodules in **4–7%**; **~10% malignant**. Thyroid cancer in pregnancy **~10 per 100,000 births**.
- **Termination not necessary**.
- **Primary therapy: partial or subtotal thyroidectomy, ideally in the 2nd trimester**. Most are well differentiated → **delay is acceptable**; aggressive tumors → prompt surgery. Then **thyroxine** replacement. ↑ VTE risk.
- **⚠ Radioiodine (I-131) contraindicated in pregnancy and lactation:**
 - Crosses the placenta → **fetal thyroid traps it → hypothyroidism**
 - Breast concentrates iodide → contaminated milk + maternal breast irradiation
- **Wait 3 months after stopping lactation** before ablation (breast involution).
- After I-131: **avoid pregnancy for 6 months to 1 year**.

5. Lymphomas

Hodgkin disease

- Probably B-cell derived; **Reed–Sternberg cells**. Common cancer in pregnancy (rising with delayed childbearing); **~1 in 12,400 births**.
- **>70%: painless supradiaphragmatic nodes** (axillary, cervical, submandibular). **~1/3 have B symptoms** (fever, night sweats, malaise, weight loss, pruritus).
- Diagnosis: **node histology**. Staging (Lugano) in pregnancy at minimum: **CXR, abdominal sonography or MR, bone marrow biopsy**. MR is excellent for thoracic and para-aortic nodes.

Table 66-3. Lugano Classification Staging System for Hodgkin and Other Lymphomas

Stage	Findings
I	Involvement in a single lymph node region or lymphoid site—e.g., spleen or thymus
II	Involvement of two or more lymph node groups on the same side of the diaphragm or cancer extends from one node group to a nearby organ
III	Involvement of lymph nodes on both sides of the diaphragm : (1) limited to spleen or splenic hilar, celiac, or portal nodes; (2) includes paraaortic, iliac, or mesenteric nodes plus those in II
IV	Extralymphatic involvement —e.g., liver, bone marrow, or lungs

Substage A = no symptoms; substage B = fever, sweats, or weight loss; substage E = extralymphatic involvement excluding liver and bone marrow; X = bulky disease.

Treatment

- Nonpregnant trend: chemotherapy for all stages.
- **Early-stage, 1st trimester: termination, or observe until >12 wk then multiagent chemotherapy**.
- **Advanced stage after the 1st trimester: ABVD** (doxorubicin, bleomycin, vinblastine, dacarbazine).
- Delaying therapy until fetal maturity is justified **only if diagnosed late** in pregnancy.
- ↑ **VTE**; **⚠ inordinate susceptibility to infection and sepsis**, even after "cure".
- **5-yr survival >90%** (same as nonpregnant). Pregnancy does not worsen the course or **trigger relapse**.

Non-Hodgkin lymphomas

- Usually B-cell (also T or NK cell). Linked to viruses: **HIV (5–10% develop lymphoma)**, EBV, hepatitis C, HHV-8.
- Infrequent in pregnancy; Lugano staging.
- **1st trimester: termination then multiagent chemotherapy**, except indolent or very early disease (observe, treat in the 2nd trimester).

- **R-CHOP** (rituximab + cyclophosphamide, doxorubicin, vincristine, prednisone): **3-yr survival 73–96%** (80 women).
- **Burkitt lymphoma** (aggressive B-cell, **EBV**): poor prognosis; **17 of 19 died within a year**.

Leukemias

- Lymphoid or myeloid; acute or chronic. **~1 in 40,000–50,000 births**.
- 72 pregnancies: **AML 44**, ALL 20, chronic 8 → **AML most common**.
- Acute leukemia: marked blood-count abnormalities, often ↑ WBC with **circulating blasts**; **diagnosis by bone marrow biopsy**.
- **Remission during pregnancy is common; termination does not improve prognosis** (may simplify care before viability; consider in early pregnancy to avoid teratogens).
- ⚠ Teratogens: **all-trans-retinoic acid (tretinoin)** → **retinoic acid syndrome**; **tyrosine kinase inhibitors**.
- Otherwise treat as nonpregnant. **AML is treated without delay**. Post-remission maintenance is mandatory; relapse → stem-cell transplant; **allogeneic transplant** → **consider early delivery**. Some chronic leukemias can wait until after delivery.
- Anticipate **infection and hemorrhage**.
- 58 cases: **75% diagnosed after the 1st trimester**; half AML with **75% remission**; only **40% live births**. APL (92 women): **91% complete remission**.

6. Malignant Melanoma

- From melanocytes, often in a **preexisting nevus**; fair-skinned women of childbearing age.
- **Biopsy** pigmented lesions that change contour/height, discolor, bleed or ulcerate.
- In some populations **the most frequent cancer in pregnancy**; **<1 to 3 per 1000 births** (under-registered, outpatient care).
- **Clinical staging**: **I** = no palpable nodes; **II** = palpable nodes; **III** = distant metastases.
- **Stage I: tumor thickness is the single most important predictor of survival**. **Clark** = 5 levels of depth; **Breslow** = thickness/size + depth.
- **Treatment**: wide local excision ± regional node dissection. **Sentinel node mapping with Tc-99m sulfur colloid** (fetal dose **0.014 mGy**) is recommended (AAD). Proposed pregnancy algorithm: **excise the primary under local anesthesia; postpone sentinel node biopsy until postpartum**.
- Chemotherapy/immunotherapy usually avoided; given if stage and prognosis demand. Distant metastases → palliative.
- **Stage for stage, survival equals nonpregnant**; but **half present at an advanced stage**.
- **Send the placenta for histology** (melanoma is the commonest placental metastasis).
- **Therapeutic abortion does not improve survival**.
- Recurrence: **60% within 2 yr, 90% by 5 yr** → **avoid pregnancy for 3–5 years**. **COCs are acceptable** contraception.

7. Gastrointestinal Tract Cancer

- **Colorectal cancer = 3rd most frequent cancer in US women**; uncommon before 40. In pregnancy **1 in 13,000 to 150,000**; rising (delayed childbearing).
- **80% arise in the rectum**.
- Symptoms: pain, distention, nausea, constipation, **rectal bleeding** → DRE, fecal occult blood, **colonoscopy** if persistent.
- **Krukenberg tumors** = ovarian metastases, often GI primary → **bleak prognosis**.

- Treatment mirrors nonpregnant: **resection** if no metastases (most are advanced). **First half: hysterectomy not needed for resection; abortion not mandated.** Later: may delay until fetal maturity unless **hemorrhage, obstruction or perforation.**
- Gastric/esophageal cancer rare; persistent unexplained upper-GI symptoms → **upper abdominal sonography ± endoscopy.**

Quick-Recall: Cancer in Pregnancy — Key Management Rules

Cancer	Frequency	Key pregnancy rule
Breast	Most common; 1/15,000	Surgery + chemo in 2nd tri ; RT after delivery ; no trastuzumab ; delay pregnancy 2–3 yr ; tamoxifen off 2 mo before conception
Cervix (invasive)	1/1200–10,000	IA1 → deliver, treat 6 wk postpartum; >20 wk non-bulky I–IIA → may delay; radical hysterectomy preferred; cesarean favored
CIN / AIS	Common	No ECC, no treatment in pregnancy ; vaginal delivery; re-evaluate ≥4–6 wk postpartum
Ovary	1/20,000–50,000	Germ cell most common; 75% early; stage surgically; CA125 unreliable
Thyroid	10/100,000	Thyroidectomy in 2nd tri or postpartum; no I-131
Hodgkin	1/12,400	1st tri early → wait until >12 wk; ABVD; survival >90%
Non-Hodgkin	Infrequent	1st tri → termination + chemo (except indolent); R-CHOP
Leukemia	1/40,000–50,000	AML treated without delay ; avoid ATRA/TKIs in 1st tri
Melanoma	<1–3/1000	Excise under local; SLNB postpartum; send placenta; delay pregnancy 3–5 yr
Colorectal	1/13,000–150,000	80% rectal; resection without hysterectomy in first half

Adnexal mass	Action
Simple <5 cm	No follow-up
5–10 cm	Surveillance if simple; resect if growing/suspicious/symptomatic
≥10 cm	Resect (14–20 wk, laparoscopic)
Suspected torsion	Surgery; untwist and conserve
Corpus luteum removed <10 wk	Progesterone support

Key Numbers to Memorize

- Cancer **1/1000–2000** pregnancies; 1/3 found prenatally; breast **39%** (INCIP)
- Radiation thresholds: malformation **0.1–0.2 Gy** (wk 2–8); intellectual disability **0.06 Gy** (8–15 wk), **0.25 Gy** (16–25 wk)
- 1st-trimester cytotoxics → **14%** of major malformations; stop chemo **3 wk** before delivery
- Ovarian transposition **3–4 cm above umbilicus**; function kept **65–94%**
- Screening: none <21; cytology q**3 yr** 21–29; co-test/primary HPV q**5 yr** ≥30. HPV prevalence **15%**
- CIN regression **61–70%**; CIN 2 → CIN 3 **7%**; residual disease after cone **43%**; transfusion **10%**
- Cervical cancer: squamous **75%**; IA1 **<3 mm**; RT vs surgery complications **30% vs 7%**; 5-yr survival **~80%**
- Trachelectomy: CL **<13 mm** at 20 wk → preterm <34 wk; survival **97%**, pregnancy **24%**, live birth **75%**
- Myomas **~2%**; abruption & breech **4x**, cesarean **6x**; vaginal birth **70%** even ≥10 cm; post-myomectomy cesarean **37 0/7–38 6/7 wk**
- Adnexal masses **1/100–2000**; malignant **1% + 1% LMP**; simple **1%** vs complex **9%** malignant
- Progesterone support if CL removed **<10 wk**: Prometrium **200–300 mg/d**; 17-OHPC **150 mg**
- Luteoma: maternal virilization **25%**, female fetus **~1/2** of these

- Ovarian cancer: **75%** early; 5-yr survival **70–90%**
- Breast: palpable **>90%**; mammogram fetal dose **0.04 mGy**, false-negative **35–40%**; node-positive **up to 60%**; metastasis $\uparrow$ **5–10%** every few months; HER2 in $\frac{1}{3}$
- BI-RADS 3 **<2%**; 4A **2–9%**; 4B **10–49%**; 4C **50–94%**; 5 **>95%**
- BRCA BSO: **BRCA1 35–40 yr, BRCA2 40–45 yr**
- Thyroid nodules **4–7%**, **10%** malignant; wait **3 mo** after lactation for I-131; avoid pregnancy **6–12 mo** after I-131
- Hodgkin **>70%** supradiaphragmatic; survival **>90%**. Burkitt **17/19** died
- Leukemia: AML **remission 75%**, live births **40%**; APL remission **91%**
- Melanoma: recurrence **60%** by 2 yr, **90%** by 5 yr $\rightarrow$ wait **3–5 yr**; Tc-99m fetal dose **0.014 mGy**
- Colorectal: **80%** rectal